

**ATTACHMENT 32 – MEMORANDUM FROM THE MASSACHUSETTS
TURNPIKE AUTHORITY (MTA) TO NTSB DATED JANUARY 12, 2007
REGARDING INSPECTION MANUAL FOR TUNNELS AND BOAT
STRUCTURES, VOLUME 5
(6 pages)**

To: Daniel Walsh, NTSB
From: Helmut Ernst, P.E., MTA Acting Chief Engineer
Date: January 12, 2007
Re: Inspection Manual for Tunnels and Boat Structures, Volume 5

I am providing the following summary and attached documents as a follow-up to yesterday's discussion regarding the Inspection Manual for Tunnels and Boat Structures, Volume 5:

A. November 2003 – July 2006

B/PB submitted the Inspection Manual for Tunnels and Boat Structures (the Manual) to the MTA in November 2003. Upon receipt of the Manual and as CA/T construction progressed, MTA continued to work with B/PB and FHWA to develop a tunnel inspection program based on the Manual. This effort included discussions regarding the Tunnel Inspection Database, a training program for inspection personnel and Bridge (Tunnel) Identification Number mapping. In late 2005, MTA took steps to initiate pilot or trial inspections based on the Manual, including engagement of an outside consultant. However, these trials were never carried out.

B. Post July 2006

Since July 2006, a number of inspection efforts have been underway throughout the MHS, including a "Stem-to-Stern" Safety Review. Findings from Phase I of this review were recently submitted to MTA and it is expected that two additional phases of review are to follow. Applicable recommendations from this three-phase review will be incorporated into future maintenance and inspection efforts. In addition, MTA has reinvigorated efforts to develop and implement a comprehensive tunnel inspection program, including:

- MTA recently issued a Tunnel Inspection and Testing Program Policy Directive that calls for routine, in-depth inspections of all tunnel elements. This policy (attached) requires the inspection of all appurtenances suspended over roadways and their hanger systems to be inspected annually and all other structural tunnel components to be inspected every 3 years. Specific inspection protocol for the MHS system will be developed and fine-tuned throughout the inspection process based on the Manual and the FHWA/FTA Highway and Rail Transit Tunnel Inspection Manual.
- In December 2006, MTA issued a Request for Proposals (attached) for a General Engineering Consultant to assist in the development of a comprehensive tunnel inspection program, including the completion of a tunnel inspection database, a training program for inspection personnel

and comprehensive inspection procedures and protocol. MTA expects to begin work on the tunnel inspection program with the selected consultant(s) this spring.

- MTA is currently reorganizing its engineering divisions to address the growing needs of the MHS more effectively. In particular, the new organization includes an MHS engineering division that will report to an Assistant Chief Engineer for the MHS. This new alignment allows for a more specialized staff to address the particular needs and complexities of the MHS. As it relates to the inspection program, this reorganization calls for a newly created tunnel and bridge inspection unit within MHS engineering with six new full-time inspection personnel to work with outside consultants to oversee and implement a rolling inspection program.
- MTA is developing a five-year capital program that will incorporate a robust tunnel inspection program, as required by the MTA Tunnel Inspection and Testing Program Policy.

Massachusetts Turnpike Authority Policy Directive

Tunnel Inspection and Testing Program

1. Purpose and Scope

- 1.1 To provide a uniform policy for tunnel inspection, life safety system testing and to establish the frequencies thereof for all tunnels under the jurisdiction of the Massachusetts Turnpike Authority.
- 1.2 The following tunnels under the jurisdiction of the MTA are included in this policy: I-90 Tunnels (I-90 connector and ramps), I-93 Tunnels (Tip O'Neill Tunnel, Dewey Square Tunnel and Ramps), Ted Williams Tunnel (TWT), Callahan Tunnel, Sumner Tunnel, City Square Tunnel (CANA), and portions of the Prudential Tunnel (includes Prudential/Copley/Hancock and proposed Columbus Center).

2. Implementation

The condition evaluation of the tunnel elements shall be consistent with the references cited below. As the tunnels listed above are each of unique design and contain unique components of varying functional life expectancies, it is acknowledged that the tunnel inspection and operation protocols mandated by this policy will evolve over time. The procedures described in the documents below are anticipated to be amended to fit the specific conditions and operational needs of each tunnel listed above.

- 2.1 Inspection Manual for Tunnels and Boat Structures, Volume 5, November 2003
- 2.2 Highway and Rail Transit Tunnel Inspection Manual, 2005. FHWA
- 2.3 MTA Operations Control Center response plan documents:
 - RP101 Tunnel Roadway Fire
 - RP101A Fire Response for I93 and I90 Tunnels
 - RP101B Fire Response for the Sumner/Callahan Tunnels
 - RP101C Fire Response for the Prudential Tunnel
 - RP101D Fire Response for CANA Tunnel

3. Frequency of Inspection

The maximum time period between inspections for tunnel components are as follows:

3.1 Ceiling panels and hanger systems over roadways supporting:

Ceiling panels, mechanical equipment (jet fans/exhaust fans), electrical equipment (lights), conduits, pipes, sign mounting assemblies, communications systems and catwalks

shall be inspected once per year.

3.2 Remaining tunnel components, including but not limited to:

Roof/lining concrete, walls, SPTC walls, columns wall panels, air supply ducts, exhaust air ducts, safety walks, barriers, curbs, roadway pavements, decks, slabs, drainage, base slab, lining concrete, pump stations, roof beams, roof beam connections, roof slabs and structural protective coatings

shall be inspected every three years.

3.3 Condition evaluation of the tunnel elements shall be based on the references cited in Section 2.1 above. Any tunnel component that has a Condition Code of 4 (poor) shall be scheduled as a priority and have the inspection frequency increased to no less than six months for components described in 3.1 and to no less than one year for components described in 3.2.

Any tunnel component that has a Condition Code of 3 (serious) or less shall be scheduled as the highest priority and have an inspection frequency increased to no less than three months for components described in 3.1 and to no less than six months for components described in 3.2.

3.4 The scheduled preventive maintenance procedures on life safety systems and fire protection testing shall be summarized and reported every six months.

3.5 Ventilation testing shall be performed on each individual tunnel and tunnel ramp ventilation zone. The tests shall evaluate the ability of the components of the ventilation system to function as designed. Additionally, the ventilation tests shall include an evaluation of the

automated and manual response plan activation from the Operations Control Center as well as local response plan activation at each vent building. Such tests shall be scheduled and conducted every six months.

4. Inspection and Testing Results

- 4.1 Data generated by the tunnel inspections shall be stored in a computerized database compatible with the National Bridge Inspection Standards.
- 4.2 Inspection results and testing results shall be utilized by MTA Maintenance and Engineering Departments to correct deficiencies, adjust scheduled preventive maintenance frequencies and plan capital improvements.

Recommended by:



Helmut Ernst, P.E.
Acting Chief Engineer

Approved by:



John Cogliano
Chairman

Effective: January 11, 2007